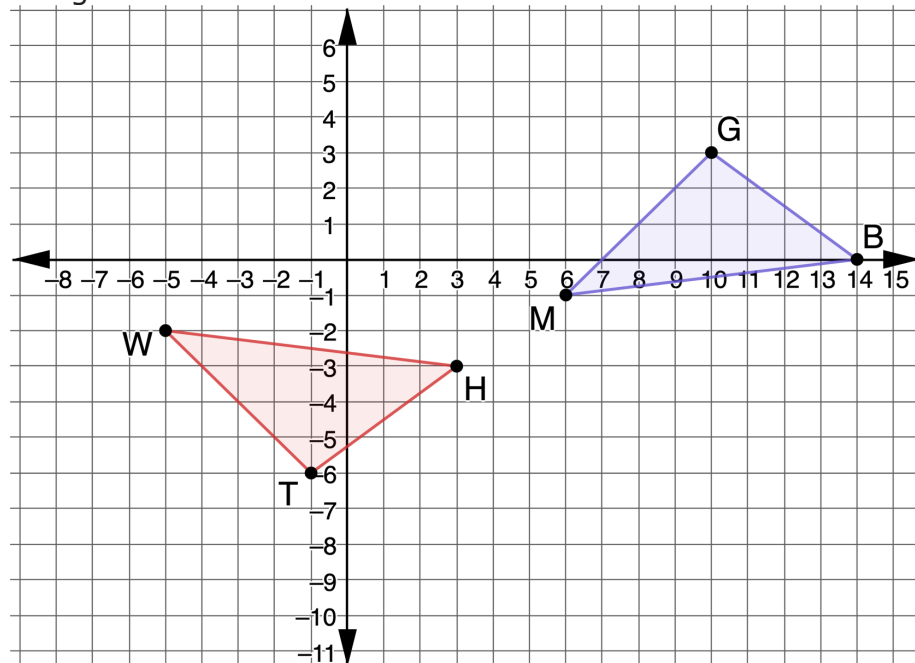




4.7.5 Wrap-Up: Ticket Out the Door

1. Triangles TWH and GBM are shown on the coordinate grid.



- a. Use written descriptions or algebraic descriptions to describe a sequence of two transformations that will map the two triangles onto each other.
- b. Complete the table that shows which vertices were mapped to each other.

$\triangle TWH$	$\triangle GBM$
T	
W	
H	

- c. Complete the congruence statement.

$$\triangle TWH \cong \underline{\hspace{2cm}}$$



Unit 4, Lesson 8: Putting It All Together



Benchmarks

MA.912.GR.2.3 Identify a sequence of transformations that will map a given figure onto itself or onto another congruent or similar figure.

MA.912.GR.2.6 Apply rigid transformations to map one figure onto another to justify that the two figures are congruent.



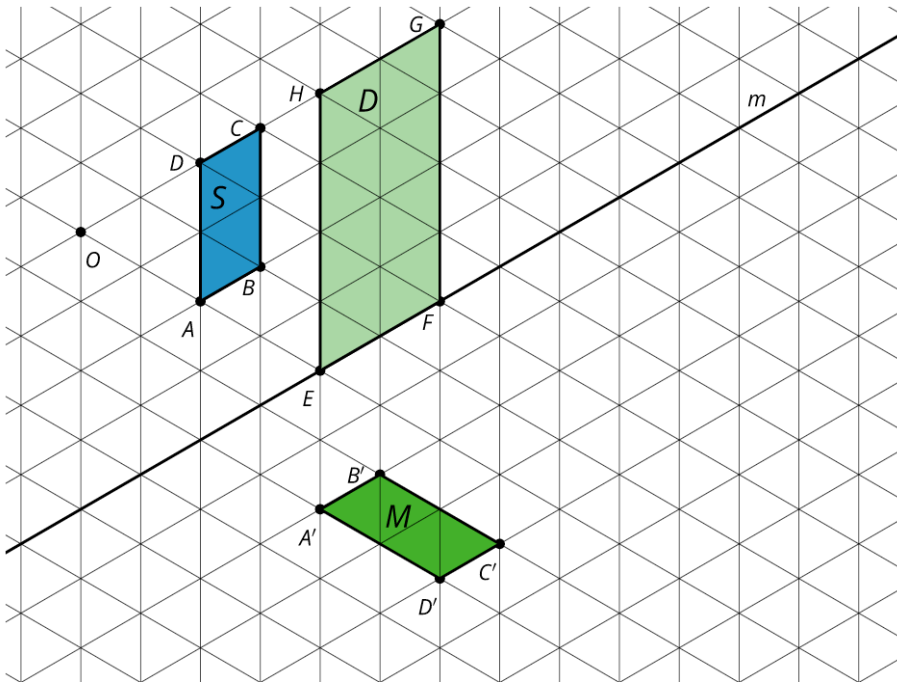
Learning Targets

- ☐ I can determine a sequence of transformations that map one figure onto another congruent figure.
- ☐ I can match the written and algebraic descriptions of a sequence of transformations given the pre-image and image of a figure.



4.8.1 Warm-Up: Notice and Wonder

1. What are three things you notice about the diagram?
2. Reflect quadrilateral $EFGH$ across line m , then translate it up 3 units. Name the resulting quadrilateral $E'F'G'H'$.
3. Check your location of quadrilateral $E'F'G'H'$ with your partner. If your locations differ, explain why both could be correct. If your locations are the same, find another possible interpretation of up 3 units and draw that quadrilateral.



4.8.2 Activity: Card Sort

Match each coordinate grid with the pre-image of $\triangle TLW$ and image $\triangle HPY$ where $\triangle TLW \cong \triangle HPY$, with the written and algebraic descriptions that map $\triangle TLW$ onto $\triangle HPY$. Use the letters in the blue, green, and orange boxes to record your matches in the table.

Coordinate Grid	Written Description	Algebraic Description
B		
D		
F		
G		
H		
M		
T		
W		
X		
Y		



4.8.3 Wrap-Up: Reflection

Draw an **X** on each line to indicate where your current learning is in relation to each target.

1. I can match the written descriptions of a sequence of transformations to the coordinate grid with the pre-image and image.

I need help.

I can't get started.

I'm getting there, but

I need more practice.

I understand this,

and I feel confident.



2. I can match the algebraic descriptions of a sequence of transformations to the coordinate grid with the pre-image and image.

I need help.

I can't get started.

I'm getting there, but

I need more practice.

I understand this,

and I feel confident.



3. I can determine a sequence of transformations that map one figure onto another congruent figure.

I need help.

I can't get started.

I'm getting there, but

I need more practice.

I understand this,

and I feel confident.



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